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The effects of forestry and agroforestry plantations on bird diversity: A global synthesis

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Abstract

The increasing expansion of productive lands around the world during the last decades constitutes a strong driver of biodiversity loss, as they are usually established near to high diversity areas. Despite many studies that have compared bird diversity between natural and productive systems, a global synthesis is still missing and important for understanding how biodiversity is being altered. We conducted a meta-analysis based on 144 case studies to assess the effects of four types of plantations (forestry, oil palm, coffee, and cacao) on bird species richness and abundance. We examined those effects in function of plantation type, latitudinal zone (temperate or tropical), geographical context (mainland or island), zoogeographic zone, and biodiversity hotspots. Plantations presented negative effects on both bird species richness and abundance. Oil palm plantations showed more negative effects followed by forestry plantations, whereas coffee and cacao agroforestry plantations had no significant effects. Those effects were geographically variable, being more pronounced in islands and temperate zones, as well as at the Oriental, Palearctic, and Neotropical zoogeographic regions, and at the Sundaland and Mediterranean Basin biodiversity hotspots. Our results showed that productive systems reduce both species richness and abundance of bird species, being insular species particularly susceptible. Exotic monocultures with low structural heterogeneity (e.g., oil palm plantations) derive in highly impoverished bird communities dominated by generalist species. We identified South East Asia, tropical South America, and the Mediterranean Basin as the most threatened regions because of the sensitivity of their bird communities and the increasing rates of native forest replacement.

KEYWORDS

abundance, commercial plantations, islands, latitude, meta-analysis, productive systems, species richness

1 | INTRODUCTION

Global economic and social development is primarily based upon biodiversity exploitation (Diaz, Fargione, Chapin, & Tilman, 2006). Within land-use changes for production purposes, we found forestry and agroforestry plantations, which are widespread across the globe, accounting for over 4 billion ha (FAO, 2018) and are expected to

increase in 13% within the next 30 years to cope with the production demand of a growing population (Sala et al., 2000). Accordingly, this increase in productive lands has resulted into the reduction of natural ecosystems (Altieri, 1999), constituting a strong driver of biodiversity loss. Since the decade of 1990, the effects of forestry and agroforestry plantations on biodiversity have been focused on specific types of productive systems based on perennial species, for instance,

forestry (e.g., *Pinus* spp. and *Eucalyptus* spp.), oil palm (*Elaeis guineensis*), coffee (*Coffea arabica*), and cacao (*Theobroma cacao*) plantations (Simonetti, Grez, & Estades, 2012), in which species diversity has been assessed. Due to the structural similarities of those productive systems to forest habitats, many of those studies compared bird diversity between those productive systems and native habitats. Those perennial plantations could be used as habitat by a subset of forest bird species, usually resulting in impoverished bird communities when compared to native forest stands (e.g., Adamack & Gruber, 2014; Azhar et al., 2011; Barlow, Gardner, et al., 2007; Castaño-Villa, Estevez, Guevara, Bohada-Murillo, & Fontúrbel, 2019; Castaño-Villa, Ramos-Valencia, & Fontúrbel, 2014; Dauber, Jones, & Stout, 2010; Estrada, Coates-Estrada, & Meritt, 1997; Faria, Laps, Baumgarten, & Cetra, 2006; Greenler & Ebersole, 2015; Hawa, Azhar, Top, & Zubaid, 2016; Srinivas & Koh, 2016). However, various studies have obtained contrasting results across different countries and productive systems, resulting in a large heterogeneity among the responses reported by those studies. Therefore, we should seek for global patterns on diversity change, aiming to understand how birds are responding to a changing world that is becoming increasingly anthropized.

Forestry and agroforestry plantations are distributed around the world according to the environmental requirements of the species planted (i.e., looking for areas with optimal conditions for the target species). Therefore, agroforestry systems have notably encroached over tropical zones, whereas forestry plantation has increased on temperate zones, following a nonrandom pattern. For example, in northern Eurasia and North America 138 million ha of native temperate forests have been replaced by forest plantations (FAO, 2015). Following the same trend, cacao plantations increase their extension over tropical Africa in ~6.4 million ha, oil palm covers approximately 15 million ha in tropical Asia, and 5 million ha are cultivated with coffee in Central America (FAO, 2018). Many of those areas experiencing major replacement of natural areas by those productive systems coincide with biodiversity hotspots, zoogeographical regions and particularly with areas of high bird diversity (Holt et al., 2013; Myers, Mittermier, Mittermier, da Fonseca, & Kent, 2000). Paradoxically, our knowledge about the effects of agroforestry systems is particularly scarce at those high-diversity areas (Jetz, Thomas, Joy, Hartmann, & Mooers, 2012). Moreover, the effects of forestry and agroforestry systems on bird diversity are expected to be more negative among those zoogeographic regions with high bird richness (Holt et al., 2013), as well as to worsen the threats on endemic and distribution-restricted species (Myers et al., 2000), or even to be more negative at highly vulnerable habitats as islands (Kier et al., 2009).

The aim of this study was to build a global synthesis about patterns of change on bird richness and abundance distribution along four forestry (trees for timber), oil palm, and agroforestry (coffee and cacao) plantations around the world. Therefore, we reviewed the literature available to perform a meta-analysis aiming to answer the following questions: (a) what is the global effect of forestry and agroforestry plantations on bird diversity? and (b) are those effects on bird richness and abundance consistent among zoogeographic zones,

biodiversity hotspots and islands? Based on the evidence available from local studies, we hypothesized that (a) bird richness and abundance will be lower in forestry and agroforestry systems compared to native forest stands, and (b) given that productive lands are non-randomly distributed around the world, the impact of forestry and agroforestry plantations will be more severe in areas of conservation concern.

2 | METHODS

2.1 | Literature survey and data inclusion criteria

We searched the available literature by topic, using the ISI Web of Science, Scopus, and ScienceDirect databases (January 1995–March 2018) using the search terms: 'cacao* + avi*', 'cacao* + bird*', 'coffee* + avi*', 'coffee* + bird*', 'oil palm* + avi*', 'oil palm* + bird*', 'plantation* + avi*', and 'plantation* + bird*.' Literature survey followed the PRISMA statement (Moher et al., 2009), which provides a standardized framework for meta-analyses and systematic reviews. Also, we followed the recommendations of Nakagawa, Noble, Senior, and Lagisz (2017) in order to get informative results by taking data independence, publication bias and the potential effects of outliers into account. Aiming to ensure a minimum quality standard of the data obtained and the replicability of the study, we limited our search to peer-reviewed articles in English, excluding grey literature and reviews (which may duplicate the information of some articles). An initial search resulted in 1,346 articles, which were limited to 309 after filtering irrelevant results (i.e., studies not focused on bird diversity or making no comparisons), which were read in full to determine if they met three criteria: (a) Comparing a native forest stand (i.e., control) with either a native or an exotic productive land (forestry or agroforestry) composed by plantation of trees, palms, or other type of cultures; (b) Reporting bird species richness or abundance for the native forest and the forestry or agroforestry plantations; (c) Reporting the mean, sample size, and a dispersion measure of the dependent variables (standard error or standard deviation). Inclusion criteria assessment was performed by one person (MBM) to avoid a potential inter-reviewer bias. As result of this screening process, we excluded 257 articles from the 309 articles initially selected as they have not met our inclusion criteria, resulting in 52 usable articles that provided 144 case studies (Figure 1). Since some papers presented more than one case study, we considered as independent cases those reporting results from different plantations.

2.2 | Effect size calculation and moderators

We used Hedges' *d* unbiased standardized mean difference to measure the effect size of each study case (Hedges & Olkin, 1985). This measurement is commonly used to estimate the magnitude of the effect of a particular situation by comparing a control and an experimental group (Gurevitch, Curtis, & Jones, 2001). We conducted separate analyses for species richness and abundance. We estimated the overall effects of forestry and agroforestry plantations on both species

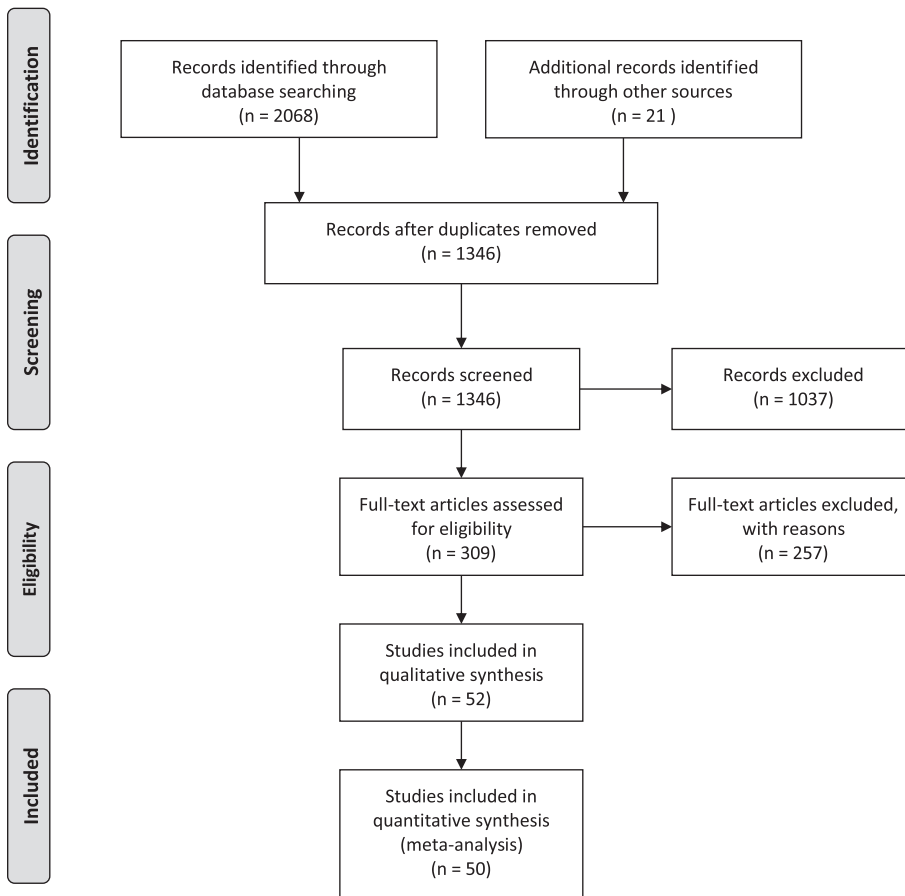


FIGURE 1 Preferred reporting items for systematic reviews and meta-analysis (PRISMA) flowchart, summarizing the sequence of information selection followed

richness and abundance. To gain further insight in our results, we used moderator variables, which are categorical variables contrasting two or more situations, to test if there are effect size differences among them and detect contrasting responses. We defined six moderator variables: (a) plantation type (forestry tree plantations, oil palm plantations, or agroforestry plantations based on cacao or coffee; please refer to Figure S1 for details on the spatial distribution of plantation types), (b) latitudinal zone (tropical or temperate), (c) geographical context (mainland or island; please refer to Figure S2 for details on their spatial distribution), (d) zoogeographic region in which the plantations occur (following Holt et al., 2013), and (e) if plantations occur within a biodiversity hotspot (i.e., areas with significant level of biodiversity or endemism; Myers et al., 2000). Given that there were strong sample size asymmetries among some moderator levels, we excluded those levels with less than four cases as we consider them little informative. Also, moderator levels with 10 or less cases should be interpreted with caution. We also extracted geographical coordinates from each study case included in order to conduct a meta-regression between effect size and latitude (as an absolute value, considering north and south as equal in terms of latitudinal variation).

2.3 | Effect model and publication bias

As we included studies reporting a variety of species, habitats, and geographic locations, we decided to use a random effects model,

which is the appropriate way to analyze our data as we do not have a single true effect for among case studies (Borenstein, Hedges, Higgins, & Rothstein, 2009). We included source paper as a random factor in the models. Therefore, we used mixed-effects models to examine effects using moderator variables. To examine the heterogeneity among moderator levels we estimated the between-group heterogeneity Q_{between} statistic, a heterogeneity measure that uses a χ^2 -distributed statistics to compare the variation between and within moderator levels (Higgins, Thompson, Deeks, & Altman, 2003). A significant Q_{between} value ($p < .05$) indicates that there is significant variation among the effects among moderator levels (i.e., moderator levels do not share a common effect, and consequently present opposite trends). Q -statistics are more appropriate for random effects models than other approaches (e.g., I^2 , τ^2) that assume a fixed-effects model structure (Borenstein et al., 2009).

To assess potential publication bias, we calculated Kendall's tau correlation with continuity correction to determine if effects and sample size were correlated. Kendall's correlation test allows us to discard potential bias in the results due to a dependence of effect size on sample size, as these measurements should be independent (i.e., not significantly correlated). Since studies reporting null effects are unlikely to be published (and consequently not included in literature surveys), we estimated the Rosenthal's fail-safe number that indicates the number of unpublished case studies with null effects needed to obtain nonsignificant effects (Hillebrand, 2008). This calculation

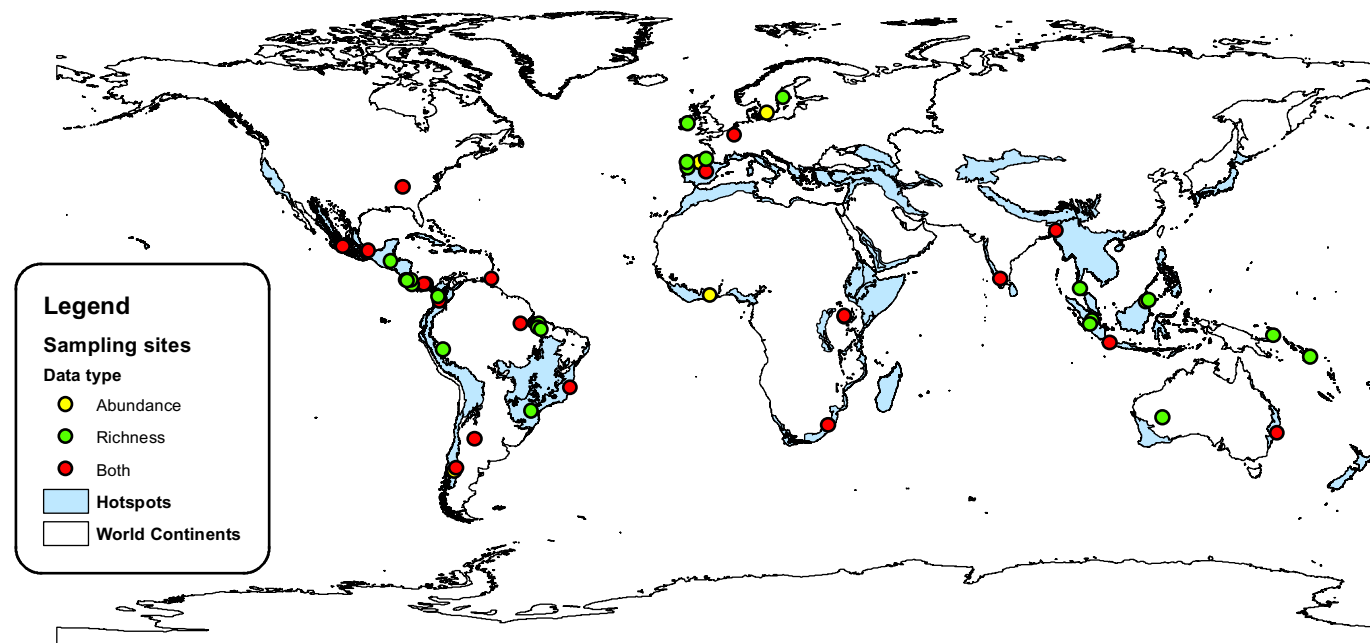


FIGURE 2 Worldwide distribution of the case studies. Biodiversity hotspots are also depicted [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/ldr.3478)]

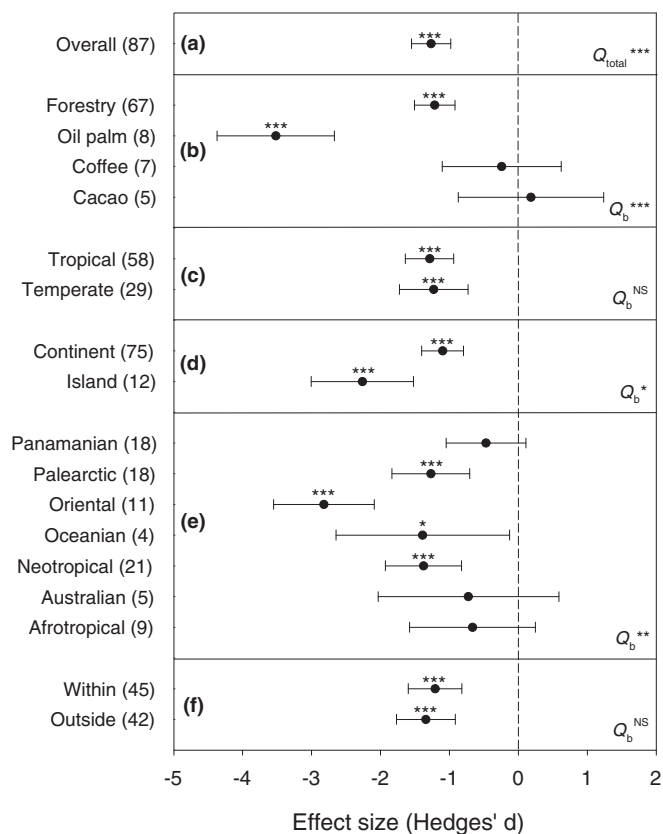


FIGURE 3 Effect of plantations on bird richness. Mean and 95% confidence intervals are shown for: (a) overall effect, (b) plantation type, (c) latitudinal zone, and (d) geographical context, (e) zoogeographic region, and (f) occurrence within biodiversity hotspots. $Q_B = Q_{between}$ statistics. Significance: $^{NS}p \geq .05$, $^*p < .05$, $^{**}p < .01$, $^{***}p < .001$

indicates a robust result when the fail-safe number is equal or greater than a threshold of $5N + 10$ (being N the number of case studies). Despite the criticism about Rosenthal's fail safe method (see Jennions,

Lortie, Rosenberg, & Rothstein, 2013), we decided to use this classic approach, as weighted approaches (such as Rosenberg's method) are known to perform poorly on random effect models (Jennions et al.,

2013). We also examined the relationship between effect and sample size by visually examining funnel plots (Hedges & Vevea, 1996); we used an Egger's test to statistically test for funnel plot asymmetry (Egger, Smith, & Minder, 1997). Then conducted a "trim and fill" analysis (Duval & Tweedie, 2000a, 2000b), which accounts for potential bias related to asymmetries of the positive and negative cases distribution, by recalculating the mean effect and confidence intervals to verify the robustness of the results (Jennions & Møller, 2002). All analyses were conducted using Comprehensive Meta-Analysis 3.0 software (Borenstein, Hedges, Higgins, & Rothstein, 2005).

3 | RESULTS

3.1 | Effects on bird diversity

From 144 case studies obtained from the literature survey, 87 were related to species richness and 57 to bird abundance (details and data sources on Table S1, available online as Supporting Information). Those case studies were distributed along the world (Figure 2). Overall, we found a negative effect of the four types of forestry and agroforestry plantations assessed (referred as plantation hereafter) on bird species richness (Figure 3a). Examining those results by plantation type ($Q_{\text{between}} = 39.63, p < .001$), we found negative effects of oil palm and tree plantations whereas cacao and coffee plantations

showed no significant effects (Figure 3b). Then, examining the results by latitudinal zone ($Q_{\text{between}} = 0.04, p = .85$), we found negative effects of plantations on bird species richness at both tropical and temperate habitats (Figure 3c). Examining results by geographical context ($Q_{\text{between}} = 8.11, p = .004$), we found negative effects of plantations at mainland sites as well as on islands, but the effects on islands were more negative (Figure 3d). Then, examining our results by zoogeographic region ($Q_{\text{between}} = 27.25, p < .001$), we found negative effects of plantations at the Neotropical, Oceanian, Oriental, and Palearctic regions, and no effect at the Afrotropical, Australian, and Panamanian regions (Figure 3e). Finally, examining the results by its occurrence within or outside biodiversity hotspots ($Q_{\text{between}} = 0.21, p = .65$), we found negative effects of plantations at both situations (Figure 3f). Examining biodiversity hotspots separately, we found that most of the negative effects of those plantations among hotspots occurred at the Mediterranean basin and Sundaland hotspots (Table S2a).

We also found a negative effect of forestry and agroforestry plantations on bird abundance (Figure 4a). Examining those results by plantation type ($Q_{\text{between}} = 0.27, p = .60$), we found a negative effects of forestry plantations whereas coffee plantations showed no effect on bird abundance (Figure 4b). Now examining the results by latitudinal zone ($Q_{\text{between}} = 7.36, p = .007$), we found negative effects of these plantations on bird abundance temperate habitats only

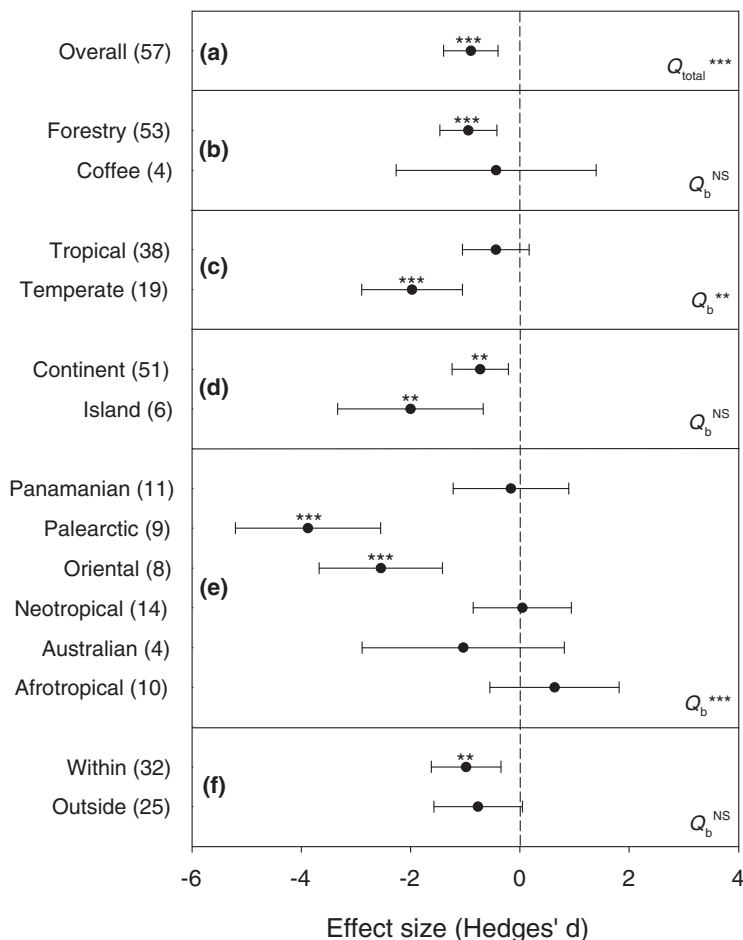


FIGURE 4 Effect of plantations on bird abundance. Mean and 95% confidence intervals are shown for: (a) overall effect, (b) plantation type, (c) latitudinal zone, (d) geographical context, (e) zoogeographic region, and (f) occurrence within biodiversity hotspots. $Q_B = Q_{\text{between}}$ statistics. Significance: ^{NS} $p \geq .05$, * $p < .05$, ** $p < .01$, *** $p < .001$

(Figure 4c). Examining results by geographical context ($Q_{\text{between}} = 3.06$, $p = .08$), we found negative effects of plantations at mainland sites as well as at islands, but the effects on islands were more negative (Figure 4d). Then, examining our results by zoogeographic region ($Q_{\text{between}} = 40.00$, $p < .001$), we found negative effects of these plantations at the Oriental and Palearctic regions only (Figure 4e). Finally, examining the results by its occurrence inside or outside biodiversity hotspots ($Q_{\text{between}} = 0.17$, $p = .68$), we found negative effects of plantations at sites within biodiversity hotspots only (Figure 4f). Examining biodiversity hotspots separately, we found that most of the negative effects of plantations among hotspots were given by the Sundaland hotspot (Table S2b).

3.2 | Latitude effect

The negative effects found for species richness were independent of the latitude (slope = 0.006, $p = .597$), the effects on bird abundance varied with the latitude (slope = 0.074, $p = .001$), indicating that the effects of forestry and agroforestry plantations on bird abundance became more negative as we move toward the poles.

3.3 | Publication bias

We found no correlation between effect and sample sizes for species richness (Kendall's tau = -0.14 , $p = .052$) or abundance (Kendall's tau = -0.08 , $p = .352$) data, indicating that there is no bias due to data dependence. Rosenthal's fail-safe number criteria indicate that the results are robust, and the potential effect of missing unpublished cases is negligible as we would need to include 9,578 additional null-effect cases to obtain nonsignificant richness results (vs. a safety threshold value of 445 cases), and to include 2,123 additional null-effect cases to obtain nonsignificant abundance results (vs. a safety threshold of 295 cases). Funnel plots showed an even distribution of positive and negative cases for richness and abundance data (Figure S3); Egger's test showed that richness funnel plot has some degree of asymmetry (intercept = -1.85 , $df = 85$, $p = .014$) whereas there was no significant asymmetry for the abundance funnel plot (intercept = -0.27 , $df = 55$, $p = .769$). The direction and significance of our results did not change after performing the "trim and fill" procedure (Table S3), indicating that they are robust and are not influenced by potential asymmetry bias.

4 | DISCUSSION

Our synthesis showed that forestry, oil palm, and agroforestry plantations negatively affect bird species richness and abundance globally, having oil palm the strongest negative effects, followed by forestry plantations. Overall, effects on species richness were more pronounced than those on bird abundance, being those effects highly variable depending on the species planted and on the geographical context in which these productive systems occur. Bird communities from islands and temperate zones showed to be more sensitive to the conversion of native forests to productive lands, including many areas of conservation concern.

A lower bird richness associated with productive systems could be related to: (a) management practices that result in a thin understory (due to burn and clear-cutting practices), (b) a reduced structural complexity (i.e., few vertical vegetation strata), and (c) the use of monocultures of exotic species (Barlow, Mestre, Gardner, & Peres, 2007; Edwards et al., 2014; Petit, Petit, Christian, & Powell, 1999; Srinivas & Koh, 2016). Particularly, bird species richness in the tropics (particularly in South America and Asia) could be severely affected by forestry and agroforestry plantations because of the close relationship that exists between bird diversity and habitat structural complexity (Castaño-Villa, Estevez, & Fontúrbel, 2014; Karr & Roth, 1971; Pearman, 2002; Terborg, 1985). On the other hand, forestry plantations in temperate habitats (e.g., *Quercus* spp., *Pinus radiata*, and *Pinus sylvestris*) showed the largest reduction in bird abundance due to the low understory heterogeneity and the large-scale loss of native forest remnants (Felton et al., 2016; Paritsis & Aizen, 2008).

The negative effects of forestry, oil palm and agroforestry plantations on bird diversity were more severe in islands (e.g., Papua New Guinea, Trinidad and Tobago, or Indonesia) rather than in mainland habitats, as our results showed. Islands usually have depauperate and fragile bird communities, which are severely affected by the replacement of native vegetation for large monocultures of exotic species (e.g., *Elaeis guineensis*, *Pinus merkusii*, *Theobroma cacao*), which are occupied by a reduced subset of native bird species (Adamack & Gruber, 2014; Paulay, 1994). Also, exotic plantations in islands could be responsible for the introduction of pests and diseases that can severely reduce local populations, or even drive them to extinction (Vitousek, D'Antonio, Loope, Rejmanek, & Westbrooks, 1997). Further, island bird populations are more susceptible to habitat loss due to their restricted distribution range and population sizes (Paulay, 1994). Therefore, biodiversity loss associated with large-scale conversion to productive lands could have catastrophic consequences in Oceania, where planted area exceeds by three times the amount of remaining native forest stands (FAO, 2018).

Bird assemblages in Asia seem to be affected by plantations on a larger extent than in other regions. In Asia, the Oriental zoogeographic region and the Sundaland biodiversity hotspot (formed by the Malay Peninsula along with Sumatra, Java, Borneo, and associated islands; Koh et al., 2013) comprise a remarkable bird diversity. Those areas were negatively impacted by forestry and agroforestry plantations, reducing both bird species richness and abundance in a greater extent than in other places. This situation is particularly critical as 45% of native forests in Southeast Asia have been converted to oil palm plantations since 1990 (Vijay, Pimm, Jenkins, & Smith, 2016), and the area covered by oil palm is expected to increase up to 74% in 2100 (Sodhi, Koh, Brook, & Ng, 2004). In this regard, low bird diversity in oil palm plantations is associated with their low structural complexity, microclimate variations, scarce understory vegetation, and intensive management practices (Azhar et al., 2013; Nájera & Simonetti, 2010; Yahya et al., 2017). Therefore, Southeast Asia could be considered amongst the most vulnerable and threatened regions in the world, as their forests and the birds that inhabit in them are facing alarming biodiversity loss and extinction risks (Brooks et al., 2006; Laurance, 2007).

Temperate forest bird communities at the Palearctic zoogeographic region and at the Mediterranean basin biodiversity hotspot (European countries and some north-African areas; Mittermier et al., 2005; Mittermeier, Turner, Larsen, Brooks, & Gascon, 2011) were negatively impacted by tree plantations, which cover approximately 83 million ha (FAO, 2015). The lowest bird diversity at Mediterranean Basin region was found at exotic tree monocultures plantations with short rotation periods, which is probably related to the limitations that management practices impose to the establishment and survival of many bird species, as they are prone to limit food resources and prey availability as well (Calviño-Cancela, 2013; Proença, Pereira, Guilhaume, & Vicente, 2010). Particularly, none of the articles included in this synthesis report the presence of any of the 32 bird species endemic to the Mediterranean basin (Mittermeier et al., 2011). Overall, endemic species occurrence and abundance is particularly low in forest plantations compared to native forests (Holbech, 2009), which could be exacerbating the effects of commercial plantations on biodiversity hotspots, but this should be confirmed with empirical data.

The Neotropical region holds the largest bird species richness on Earth, with an important number of endemic and migratory species, which are increasingly threatened by the expansion of coffee and oil palm plantations (Gilroy et al., 2015; Philpott et al., 2008). Both coffee and oil palm plantations have been associated with extensive loss, fragmentation, and degradation of the native forests as result of a rapid and irreversible land use change (Durães, Carrasco, Smith, & Karubian, 2013; Grau & Aide, 2008). However, contrarily to what was found for oil palm, coffee, and cacao plantations (which are usually grown as agroforestry systems) at the Panamanian region seem to have neutral effects on bird diversity, as they offer shaded understory conditions, serve as corridors between native remnants across the landscape, and occasionally are cultivated associated with native tree species (Harvey & Villalobos, 2007; Schroth & Harvey, 2007).

Examining the habitat loss and transformation issue beyond the perennial plantations examined here, the global change scenario is even more critical if we also consider the expansion of annual crops such as maize (*Zea mays* L.), rice (*Oryza sativa* L.), and soybean (*Glycine max* (L.) Merr.), which cover ca. 487 million ha (FAO, 2018). Such crops also have negative effects on the associated biodiversity (Azman et al., 2011; Estrada, Coates-Estrada, & Meritt, 1997; Newbold et al., 2015; Wilson et al., 2017) and usually are dominated by a few generalist bird species (Di Giacomo & López de Casenave, 2010). As happens with the productive lands assessed in this meta-analysis, annual crops are also experiencing a rapid expansion on developing countries that hold a large biodiversity (e.g., Brazil, Argentina, and India are among the top 10 soybean producers in the world; FAO, 2018).

5 | CONCLUDING REMARKS

Our results show that forestry and agroforestry plantations have negative effects on bird species richness and abundance worldwide, being variable across the plantation types, zoogeographic regions, and latitudes. Avifauna communities of Southeast Asia, Europe (Mediterranean Basin), and

South American tropical forests are particularly vulnerable to biodiversity loss as plantations are encroaching on native forests. This situation could be even worst in developing countries (e.g., Brazil, Colombia, Ghana, Guatemala, India, Indonesia, Kenya, Papua New Guinea, South-Africa, Perú, and Solomon Islands), which hold a large biodiversity but are facing economies in rapid expansion, based on the exploitation of natural resources, which may lead to an irremediable biodiversity loss in the next decades.

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DATA AVAILABILITY STATEMENT

Original data associated with this article can be accessed at the figshare repository: <https://doi.org/10.6084/m9.figshare.6984797>.

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